Simple Algebra

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

A) Easy: Solve: 5x − 8 = 27.
Method & answer: 5x = 35, so x = 7.

B) Standard: Solve: 3(2x − 1) = 4x + 9.
Method & answer: 6x − 3 = 4x + 9, so 2x = 12, x = 6.

C) Advanced: Two numbers have a sum of 40 and a difference of 12. Find the two numbers.
Method & answer: (40+12)/2 = 26 and 40−26 = 14. The numbers are 26 and 14.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 6x − 5 = 37.

2. Solve: 4x + 9 = 3x + 20.

3. Solve: 3(x + 4) = 27. A) 5 B) 6 C) 7 D) 8

4. Solve: 2x/3 = 8.

5. Solve: 5(x − 2) = 3x + 4.

6. Select ALL of the equations below where x = 6: A) 4x − 3 = 21 B) 2(x + 1) = 16 C) 5x + 2 = 34 D) 3x − 4 = 14 E) 7x − 1 = 41

7. Solve: 7x + 3 = 5x + 21.

8. Solve: (x − 5)/2 = 6.

9. Solve: 4(2x − 3) = 44. A) 6 B) 7 C) 8 D) 9

10. Two numbers have a sum of 50 and a difference of 8. Find the larger number.

11. Solve: 9x − 4 = 6x + 17.

12. Select ALL of these equations where x = 9: A) 3x − 7 = 20 B) 2x + 5 = 23 C) x/3 + 2 = 5 D) 5x − 10 = 35 E) 4x + 1 = 30

13. Solve: 6(x + 3) − 2x = 34.

14. Solve: 3x + 5 = 2x + 5 + 12. A) 8 B) 10 C) 12 D) 14

15. Solve: 10x − 3(x + 2) = 29.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 2(x + 5) = 3(x − 1).

2. Solve: (2x + 4)/3 = 6.

3. Solve: 5(x − 2) = 2(x + 7). A) 6 B) 7 C) 8 D) 9

6. Select ALL of the equations below where x = 8: A) 3x − 4 = 20 B) 2(x + 1) = 20 C) 5x − 10 = 30 D) 4x + 3 = 35 E) 6x − 2 = 46

7. A number is trebled, then 7 is subtracted, giving 26. Find the number.

8. Solve: 3(2x + 1) − 5 = 34.

11. Select ALL of these equations where x = 5: A) 3(x+2) = 21 B) 2x + 15 = 25 C) 4x − 5 = 20 D) (x+3)/2 = 4 E) 6x − 10 = 20

12. A rectangle's length is twice its width. If the perimeter is 54cm, find the width.

13. The nth term of a sequence is 3n + 1, and separately x satisfies 3x + 1

= 40. Find x.

4. The sum of three consecutive numbers is 72. Find the smallest number.

9. Two numbers have a sum of 60 and one is 3 times the other. Which is the smaller number? A) 12 B) 15 C) 18 D) 20

14. Solve: 7x − 2(x + 3) = 29. A) 6 B) 7 C) 8 D) 9

5. Solve: 4x + 3(x − 2) = 36.

10. Solve: (x + 6)/4 = x − 3.

15. A number doubled and increased by 11 equals the number trebled decreased by 4. Find the number.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 5(x − 3) = 2(x + 6).

6. A number is doubled, 7 is subtracted, and the result is trebled to give 45. Find the original number.

11. A number multiplied by itself gives the same result as the number plus 20. Find the positive value of the number.

2. Two numbers have a sum of 58 and a difference of 14. Which is the larger number? A) 22 B) 30 C) 36 D) 44

7. Solve: 6x − 3(x − 4) = 2x + 20. A) 6 B) 7 C) 8 D) 9

12. A father is 4 times as old as his son. In 6 years he will be 3 times as old as his son. How old is the son now?

3. A rectangle's length is 4cm more than 3 times its width. If the perimeter is 64cm, find the width.

8. The sum of two numbers is 48. One number is 9 more than twice the other. Find the smaller number.

13. Solve: 8x − 5(x − 2) = 4x + 1. A) 6 B) 7 C) 8 D) 9

4. Select ALL of the equations below where x = −3: A) 2x + 9 = 3 B) 5x + 4 = −11 C) 3(x + 5) = 6 D) 4x − 1 = −13 E) x/3 + 5 = 6

9. A taxi fare is £4 plus £2.50 per mile. Another taxi charges a flat £16.50 regardless of distance. After how many miles do the two fares become equal?

14. The perimeter of an isosceles triangle is 30cm. The two equal sides are each 3cm longer than the base. Find the base length.

5. Three consecutive odd numbers add up to 81. Find the largest number.

10. Select ALL of the values of x below that make 3x − 7 greater than 20: A) 8 B) 9 C) 10 D) 11 E) 12

15. Two consecutive even numbers multiply to give 168. Find the larger number.